Frailty Syndrome in Older Adults: Modeling as a Complex Adaptive System

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Abstract – Frailty syndrome is a common health condition that makes patients more vulnerable, existing mostly in older adults, where it tends to coexist with several chronic diseases such as heart disease, diabetes, and arthritis. More than just a clinical state, frailty is a complex adaptive system where it emerges from many different interacting factors, making it highly dynamic. This project aims to treat frailty as a complex adaptive system, examining interconnected factors and their behaviors to better understand how each component affects the overall system. The interactions between connected nodes (weakness, slowness, low activity and exhaustion) in a network model is the key idea that emerges the complexity of the frailty system. Emergence was demonstrated through cascade propagation and nonlinear threshold effects. Self-organization was proven due to complexity metrics and the existence of attractors of convergence and resilience.

I. Introduction

As the population ages, frailty is becoming increasingly common. It is specifically recognizable by multiple illness conditions coexisting. With underlying disease, elders can easily have another disease related to the ones they are having. Once the body has reached its endurance, or the point where frailty occurs, the body may easily collapse. Frailty can lead to higher risk of fall, hospitalization or worse, mortality. From previous approaches, frailty was usually considered to build up from different single deficits. That approach might not be tight enough to capture the whole system's behavior when it cannot track every organ or illness at once. It is crucial to have a complexity approach to view frailty as an entire system, which is able to capture the smallest properties, and how the system will adapt to endless emergencies.

Previous network models [1], [2] have established the mathematical framework for damage propagation but primarily focused on large scale networks with dozens of nodes. Whether the same emergent and self-organizing dynamics hold in a smaller network using only the core Fried *et al.* [3] phenotype criteria remain unexplored. This project addresses these gaps by implementing a simplified 4-node model to isolate the fundamental mechanisms of emergence and self-organization in frailty.

This research implemented the network model from Mitnitski *et al.* [1], with phenotypes from Fried *et al.* [3] with six experiments conducted to gain a deeper understanding of the system emergence and self-organization. Based on that, this study investigates two hypotheses: (H1) frailty emerges from the internal interactions between the factors within the network, where a single state of illness can non-linearly affect the whole system; and (H2) the system self-organizes toward stable health states driven solely by internal dynamics, without requiring external intervention.

II. Related works

A. Network Models of Aging and Frailty

Mitnitski *et al.* [1] contributed to the foundational knowledge of the frailty network. The suggestion of viewing frailty as a binary stochastic network, each deficit is a node. Their approach has shown the network dynamics with time-independent variables.

Farrell *et al.* [2] expanded the model to population level with realistic mortality rate for parameter adaptation. García-Peña *et al.* [4] confirmed using empirical data that frailty behaves as a complex network, where complexity increases as edges, nodes and deficits grow.

B. Frailty Phenotype and Nonlinear Risk

The framework from Fried *et al.* [3] has been widely used to classify frailty in clinical settings, where it defines five main phenotypes: unintentional weight loss, self-reported exhaustion, weak grip strength, slow walking speed and low physical activity.

Ghachem *et al.* [5] proved that the frailty risk does not emerge linearly. When the number of deficits has reached a specific threshold level, the system becomes exponential and hardly recoverable.

C. Complexity and Resilience Metrics

Gijzel *et al.* [6] found that frail older adults exhibited significantly higher variance and stronger cross-correlations across physical, mental, and social health domains compared to non-frail individuals. This suggests that as the system approaches the frail state, dynamic fluctuations increase — reflecting a loss of stable regulatory capacity and reduced systemic resilience.

III. Methodology

A. Network Structure

The model was implemented based on the framework of Mitnitski *et al.* [1], using the network interacting nodes and the key criteria were from Fried *et al.* [3] frailty phenotype. Originally, the model contained five factors, but due to the

limitations of the dataset, and the time-dependent parameters were not included as intended [1], [2], at the same time, it did not affect the model since the act of switching states between the nodes was the focus point [1]. As a result, the model was simplified into four nodes representing the frailty phenotype: weakness, slowness, low activity and exhaustion.

According to the cycle of frailty from Fried *et al.* [3], the connections between every node i were defined by their corresponding edges. Each node holds the value k_i that defines the number of nodes that have the direct influence on that single node.

Every node has two states which were defined by using the value $d_i = 0$ for healthy nodes, and $d_i = 1$ for damaged nodes. At each time step, transitions between states are stochastic [1], [2].

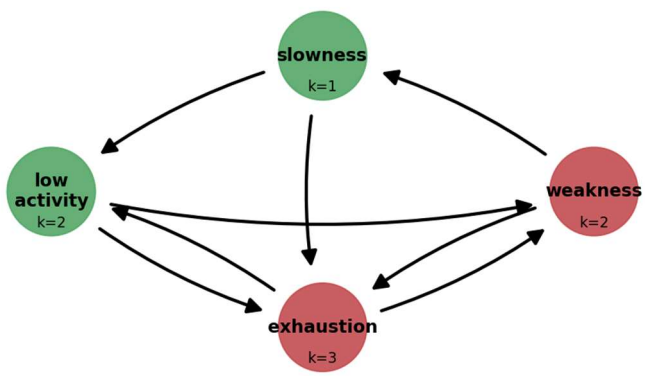

Fig. 1. Frailty network structure. Red node: damaged; Green node: healthy.

B. Mathematical Model

1) Local frailty

Each node in the network has its own local values defining their properties and behaviors. Local frailty is a dynamical variable, changing depending on how dense the node was connected. As a result, local frailty defines the local damage and repair rate of a node depending on the initial state of neighbor nodes [2]:

$$f_i = \sum_{j}^{connected} d_j / k_i \,, \tag{1}$$

where f_i is the local frailty of node j defined by its damage state and in-degree nodes k_i [1].

2) Damage rate and repair rate

The damage rate Γ_+ decides the chance of a single node affecting its neighbor nodes when the repair rate Γ_- decides the chance of that node recovering from the damage [1], [2]:

$$\Gamma_+ = \Gamma_0 \exp(\gamma_+ f_i), \quad \Gamma_- = \frac{\Gamma_0}{R} \exp(-\gamma_- f_i). \tag{2}$$

The γ_+ parameter is strongly important, leading to the exponential damage arising within the network, when the γ_- and the repair resistance R are not much significant in drawing the state back to undamaged [1], [2].

3) Frailty index (FI)

The frailty index (*FI*) provides a quantitative measure of an individual's health status by calculating the proportion of accumulated age-related deficits [1], [2]. Higher *FI* reflects greater deficit accumulation and is strongly associated with adverse health outcomes.

In the network model, the *FI* at each time step is computed directly from node states as the fraction of damaged nodes over total nodes [1]:

$$FI = \sum_{i} d_i / n, \tag{4}$$

where d_i is the state of node i and $n = 4$ is the total number of nodes. Since the model uses four discrete nodes, $FI \in \{0,\ 0.25,\ 0.5,\ 0.75,\ 1.0\}$. Following Fried *et al.* [3], individuals are classified as robust ($FI = 0$), pre-frail ($FI = 0.25$–0.50, indicating 1–2 damaged criteria), or frail ($FI \geq 0.75$, indicating 3 or more damaged criteria).

C. Model parameters

The original parameters reported by Farrell *et al.* [2]: $\Gamma_0 = 0.00113/year$, $\gamma_+ = 10.27$, $\gamma_- = 6.5$, $R = 1.5$ – were designed for a continuous-time algorithm and for a larger scale network. To adapt for the present discrete-time, 4-node implementation, parameters were first bounded by mathematical constraints, then iteratively verified to preserve qualitatively distinct dynamics across frailty groups. All parameters were scaled down to: $\Gamma_0 = 0.008/\text{step}$, $\gamma_+ = 2.0$, $\gamma_- = 0.3$, those reductions are further supported by Farrell *et al.*[2]. Repair resistance R was varied per frailty group following Gijzel *et al.* [6]: $R = 0.25$ (robust), $R = 0.50$ (pre-frail), and $R = 1.50$ (frail), where lower R reflects stronger repair capacity and higher systemic resilience.

D. Dataset

The dataset consists of $n = 200$ subjects classified into three frailty groups: robust ($n = 92$), pre-frail ($n = 90$), and frail ($n = 18$). Clinical measures were synthetically generated based on Fried *et al.* [3], phenotype criteria and binarized using the following thresholds: weakness if grip strength < 20 kg, slowness if gait speed < 0.8 m/s, low activity if energy expenditure < 270 kcal/week, and exhaustion as self-reported. Synthesized values show approximately ±10–15% deviation from published clinical reference means. Initial node states per group were derived by majority vote – a node begins damaged if over 50% of subjects in that group meet the corresponding criterion. Due to the small frail group size ($n = 18$), this estimate is less stable than for the other two groups.

IV. Experiments

A. Emergence Experiments

E1 – Cascade Propagation: Each of the four nodes was initialized as the sole damaged node while all remaining nodes stayed healthy ($FI = 0.25$), with an all-healthy control ($FI = 0$) included for baseline. *FI* trajectory was recorded over 200 steps across 50 Monte Carlo runs to examine whether damage propagates through network interactions alone [1]. Fifty runs ensure stable mean estimates given the stochastic transitions,

and 200 steps allow sufficient time for the system to reach steady state.

E2 – Nonlinear Threshold: The number of initially damaged nodes was varied from 0 to 4, testing all possible node combinations at each level. Each combination was repeated across 30 independent runs for 200 steps and mean final *FI* per damage count was recorded to identify any threshold effects [5].

E3 – Network Versus Isolated: Two conditions were compared starting from the same initial state (2 damaged nodes): connected, where nodes interact normally ($\gamma_+ = 2.0$), and isolated, where neighbor coupling was removed ($\gamma_+ = 0$). Both ran for 200 steps across 50 independent runs, with final *FI* distributions compared between conditions[1], [5]. This isolates network coupling as the sole variable, directly testing whether emergence requires interaction.

B. Self-Organization Experiments

SO1 – Attractor Convergence: Thirty random binary initial states were generated and each evolved over 300 steps using default parameters ($R = 0.5$). The distribution of final *FI* values was recorded to check whether the system consistently converges to stable attractors regardless of starting condition [1]. Random initialization ensures the attractor landscape is explored without bias toward any particular state.

SO2 – Resilience and Attractor: Starting from the same pre-frail state (weakness = 1, low activity = 1, FI = 0.5), repair resistance R was varied across six values spanning from high resilience (R = 0.2) to low resilience (R = 3.0). Each condition ran for 300 steps across 50 independent runs, with mean final FI recorded per R value [1], [6].

SO3 – Complexity Metrics: For each frailty group, 50 simulation runs of 500 steps were conducted using group-specific *R* values (robust $R = 0.25$, pre-frail $R = 0.50$, frail $R = 1.50$). The final 200 of 500 steps were used to ensure metrics reflect steady-state behavior, not transient dynamics. Three metrics were be captured: total state transitions, *FI* variance, and mean *FI* [6].

V. Discussion

A. Emergence Findings

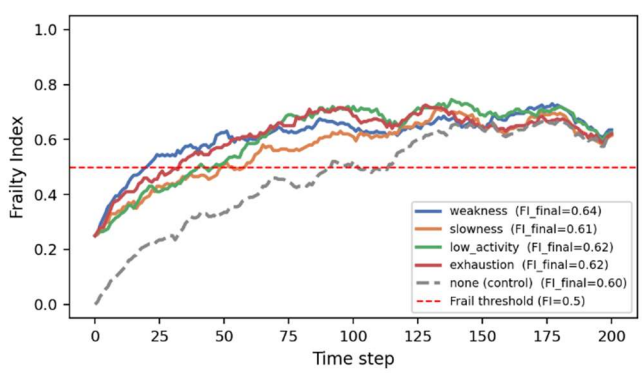

Fig. 2. E1: Cascade Propagation from Single Node Damage

The results in Fig. 2 show that even when only one node is initially damaged ($FI = 0.25$), the entire system crosses the frail threshold ($FI > 0.5$) within 25-50 steps, converging to final *FI* ~ 0.61–0.64. This demonstrates that damage is not contained within a single node but propagates through the network via cascade mechanism due to exponential coupling in the damage rate. This supports H1 – frailty emerges from interactions between nodes rather than from any single deficit.

The stacked bar in Fig. 3 shows a tipping point at $n = 2$ initially damaged nodes – here the distribution shifts notably toward $FI = 0.75$ and $FI = 1.0$ attractors, while $n = 0$–1 still maintains moderate *FI* levels. The relationship between initial damage count and final *FI* is nonlinear – once two or more nodes are simultaneously compromised, the network's repair capacity is overwhelmed and the system tips. This validates H1 through the nonlinear threshold effect from Ghachem *et al.* [5].

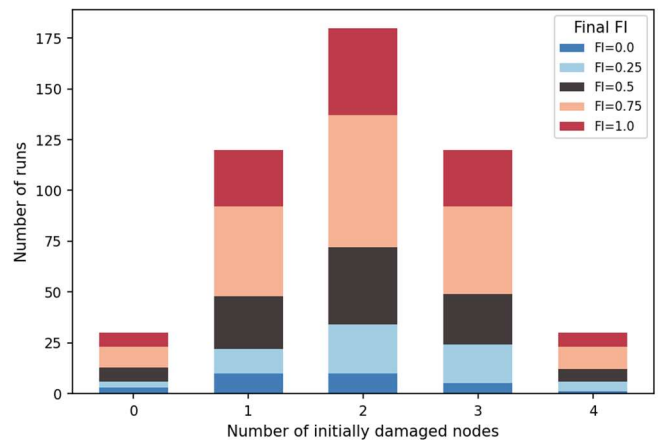

Fig. 3. E2: Final FI Distribution per Initial Damage Count

In Fig. 4, when $\gamma_+ = 0$ (isolated), nodes cannot accelerate each other's damage, so *FI* fluctuates around 0.3–0.4, staying below the frail threshold. In contrast, the connected network with the same initial state (2 damaged nodes) drives *FI* above 0.6. This demonstrates frailty does not reside in any single node but emerges from the interaction structure of the network, directly supporting H1.

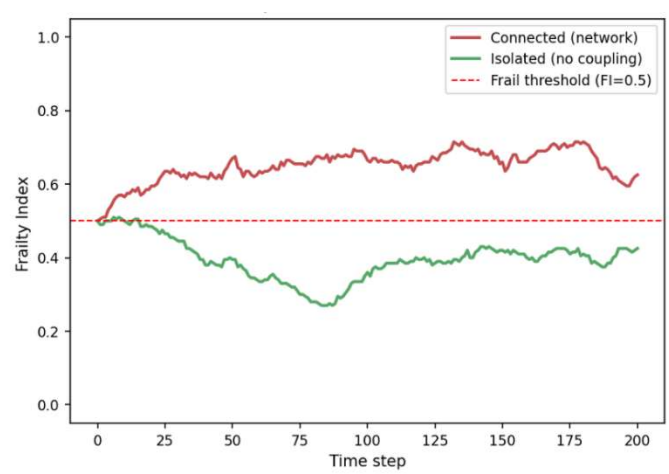

Fig. 4. E3: FI Trajectories: Connected versus Isolated Nodes

B. Self-organization Findings

From 30 random initial states, Fig. 5 demonstrates the system self-converges to discrete attractors with the majority (> 60%)

converging to $FI \geq 0.5$. Without external intervention or time-dependent parameters, the system self-organizes toward stable states through positive feedback in damage or repair rates. This supports H2 – self-organization toward stable health states driven solely by internal dynamics.

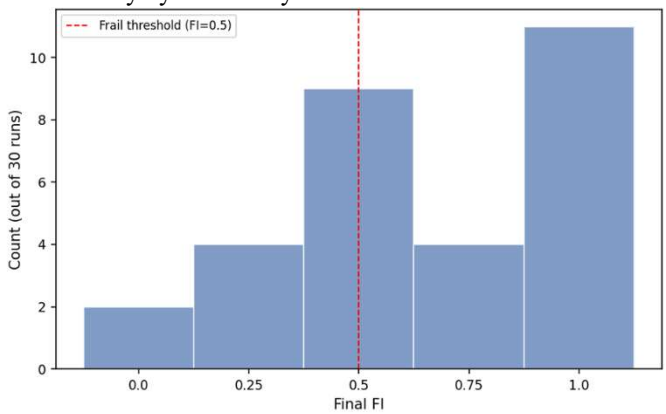

Fig. 5. SO1: Distribution of Final States (Attractors)

The graph in Fig. 6 shows a bifurcation point between R = 0.2 and R = 0.35 — at R = 0.2 the system self-organizes to robust ($FI \sim 0.28$), but from $R \geq 0.35$ the system tips to the frail attractor and stays there (FI > 0.5). The relationship is nonlinear: a small decrease in resilience near the critical point causes catastrophic effect on final FI. This supports H2 – repair resistance R is the control parameter determining which attractor the system self-organizes toward, consistent with Gijzel *et al.* [6] on resilience capacity.

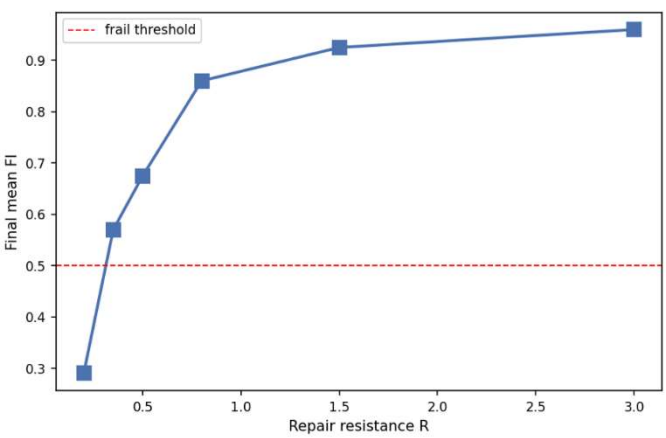

Fig. 6. SO2: Self-Organized Stable State versus Repair Resistance

In Fig. 7 the robust group has median ~47 state transitions, while the frail group has only ~15 transitions. This shows that robust systems actively cycle between damage and repair (high dynamism), whereas frail systems are stuck in the damaged state with low transitions. Although Gijzel *et al.* [6] found that frail individuals exhibit higher variance (due to critical slowing down before attractor lock-in), the 4-node binary model creates a frail attractor too strong for nodes to fluctuate – the low variance and low transitions reflect rigid, locked-in behavior once the system reaches the frail state. This supports H2 regarding self-organized stable states driven by internal dynamics.

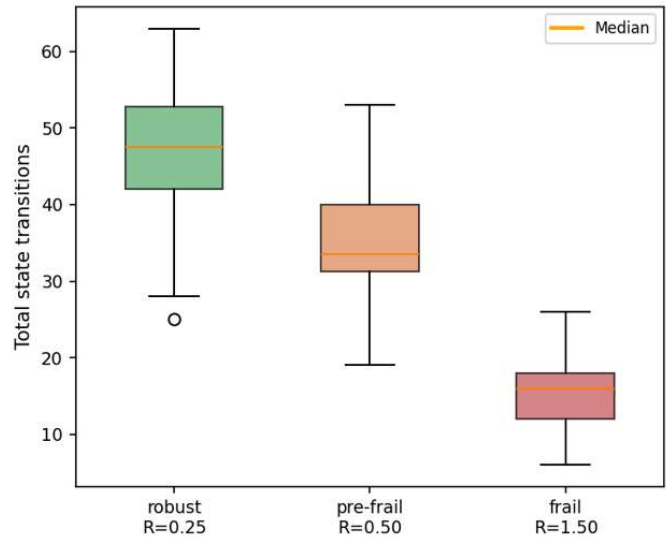

Fig. 7. SO3: State Transitions Across Frailty Groups

C. *Limitations*

First, the dataset used is synthetic, generated from published Fried *et al.* [3] phenotype thresholds rather than real clinical measurements, which may not fully capture the variability found in real patient populations.

Second, the model is currently limited to four core frailty variables, excluding comorbidities such as heart disease and diabetes that are known to influence frailty progression.

Third, edge weights and resilience parameters were set based on literature estimates rather than empirically derived values.

For future work, incorporating real longitudinal data, expanding the network to include comorbidity nodes, and validating model outputs against clinical outcomes would significantly improve the model's reliability and applicability.

VI. Conclusion

From the complexity perspective, the frailty system shows a close relationship between nodes, where emergence only arises from interconnected factors, showing that every behavior matters. The three emergence experiments (E1, E2, E3) supported (H1) by demonstrating cascade propagation from single node damage, nonlinear threshold effects at two damaged nodes, and the critical role of network coupling in driving frailty. The three self-organization experiments (SO1, SO2, SO3) supported (H2) by showing attractor convergence from random initial states, bifurcation behavior controlled by repair resistance R, and distinct complexity metrics across frailty groups.

Experiments demonstrated that frailty emerges at the system level even when no single node reaches the impairment threshold individually, and the system self-organizes into three stable states: robust, pre-frail, and frail, driven solely by local interactions. The system can converge to attractors, making it stable over time, which is a key concept for further studies on frailty syndrome. With better optimizations, this model can contribute to early detection and intervention strategies for frailty progression in aging populations.

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